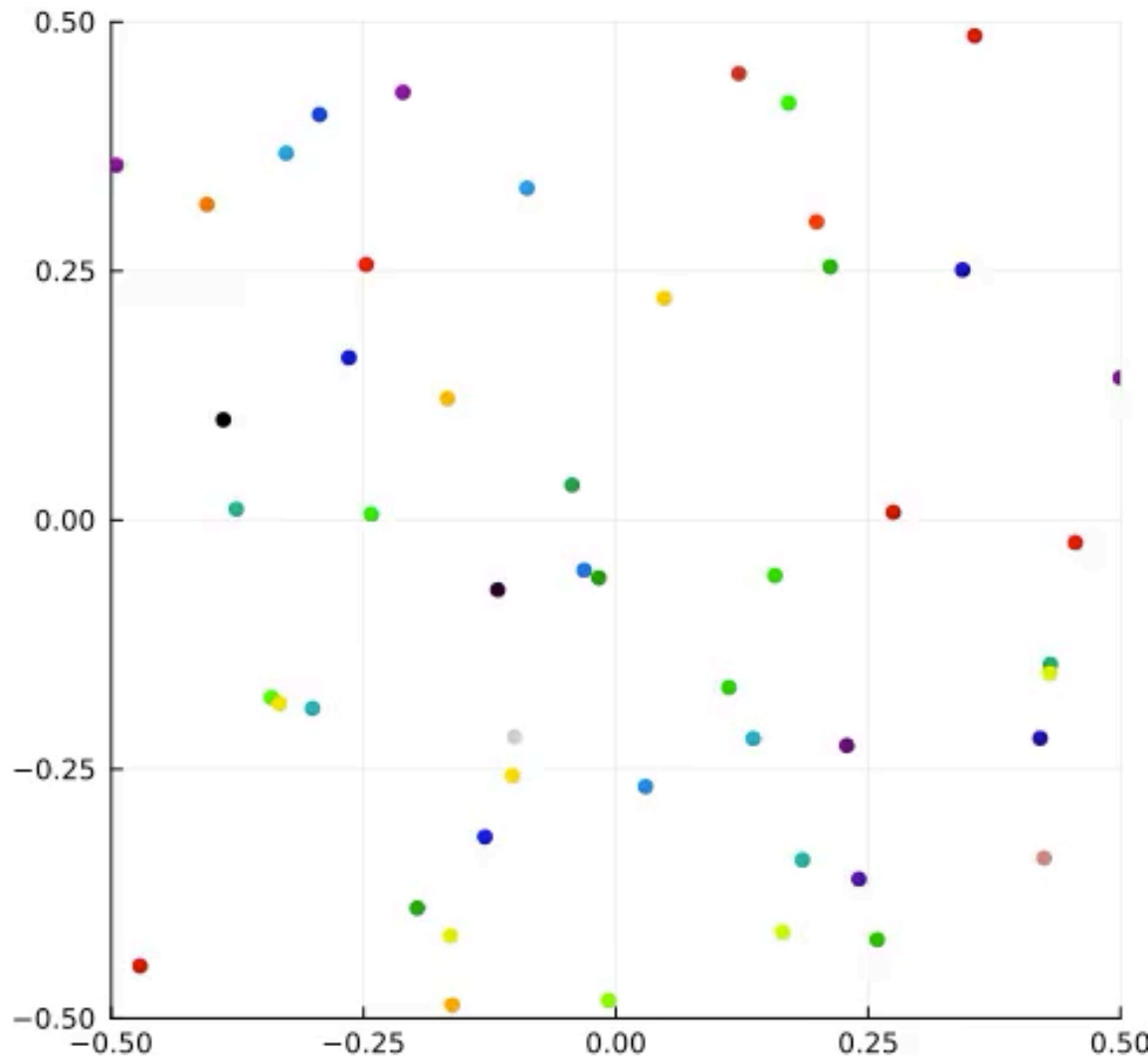
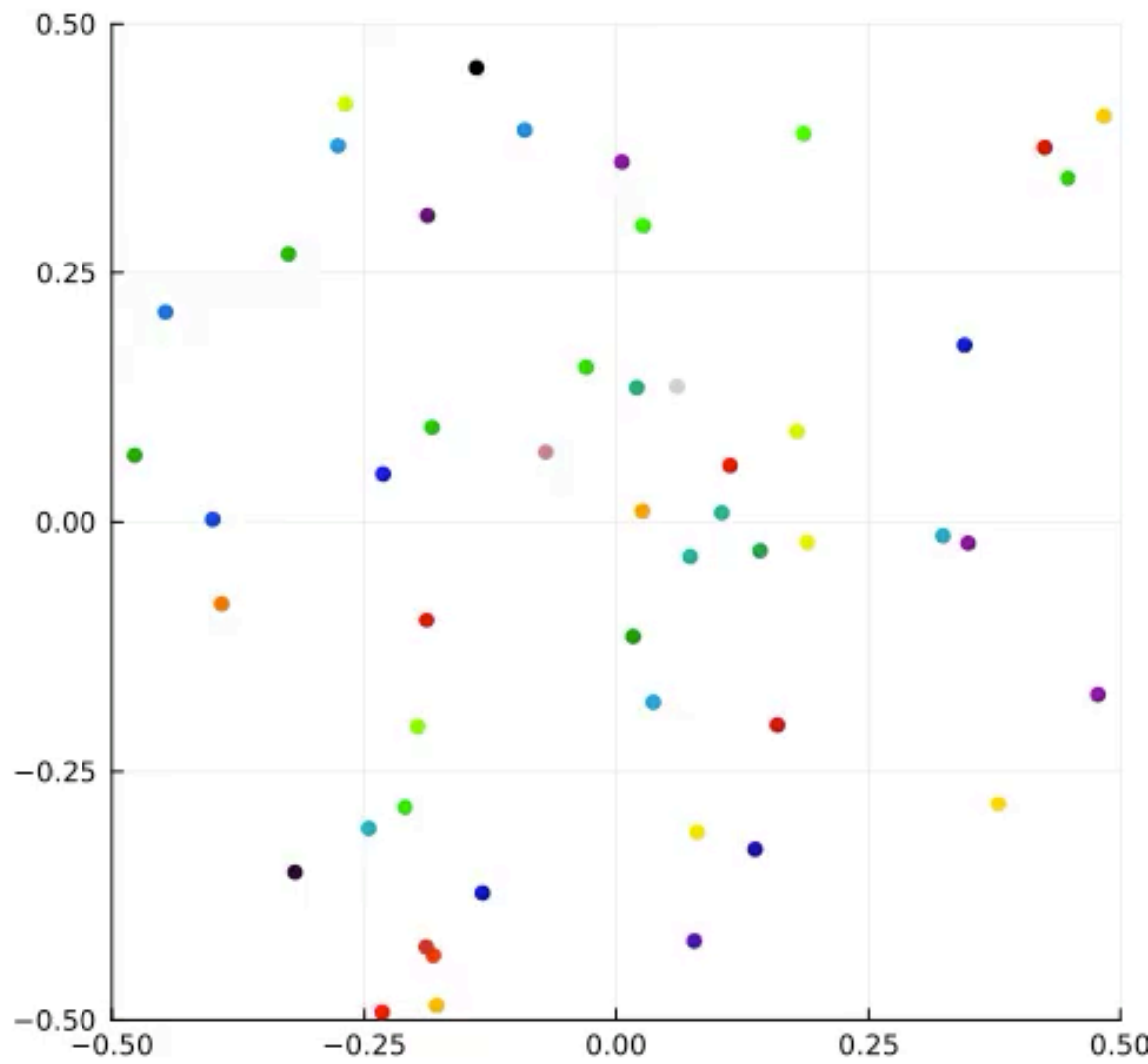


$t=0.0$



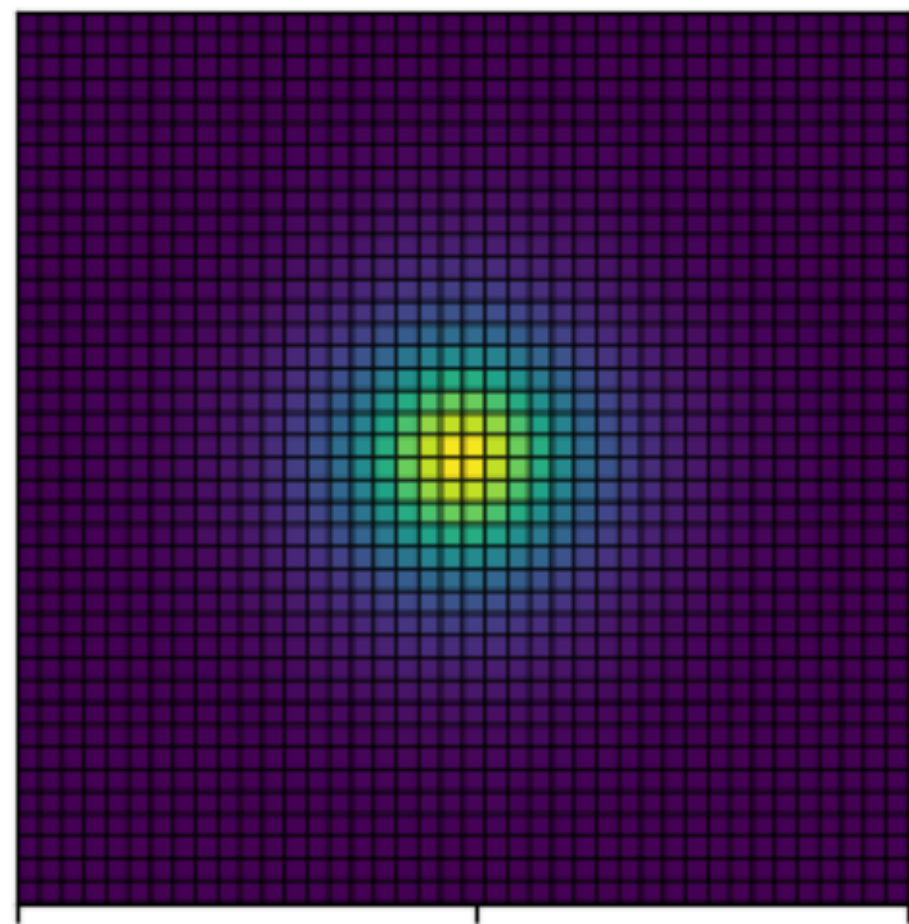
$$\tau = 0$$

$t=0.0$



$$\tau = 0.15$$

$t = 3$



ρ

12.5

10.0

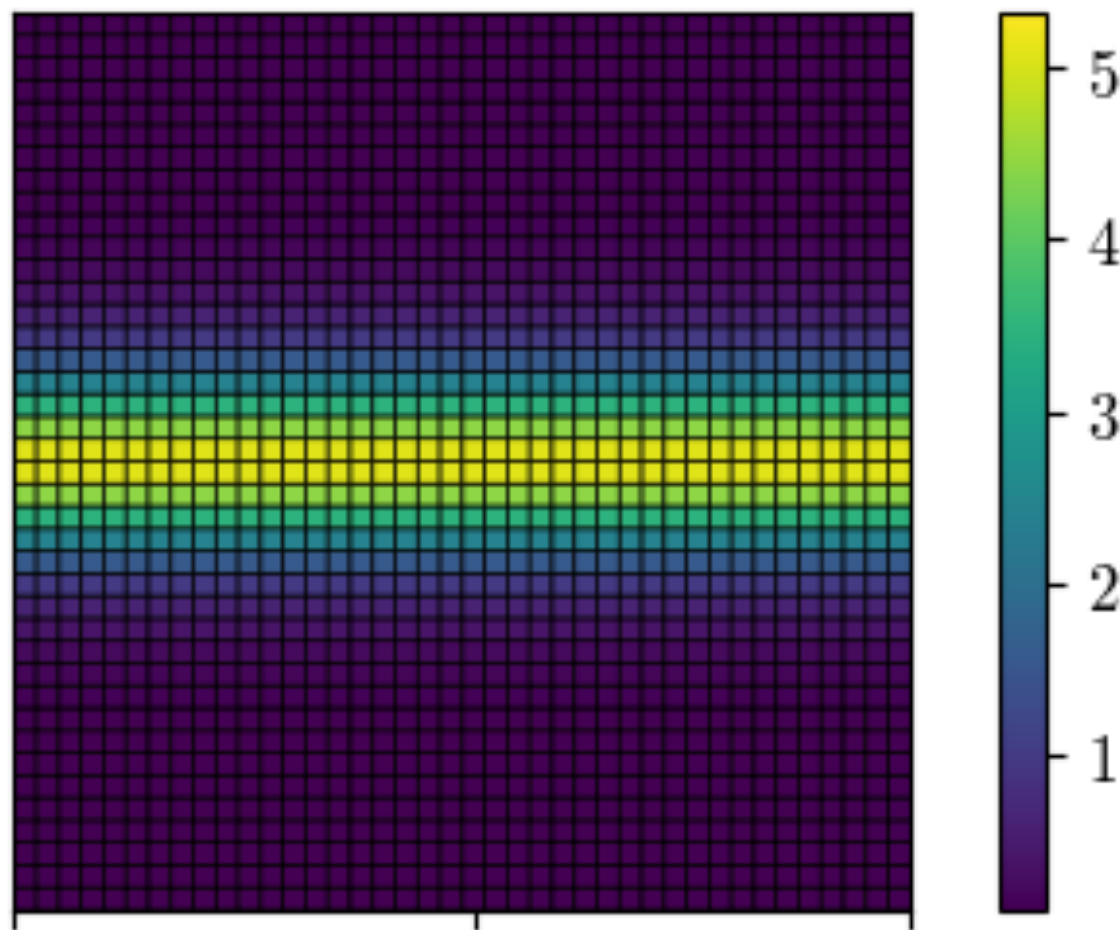
7.5

5.0

2.5

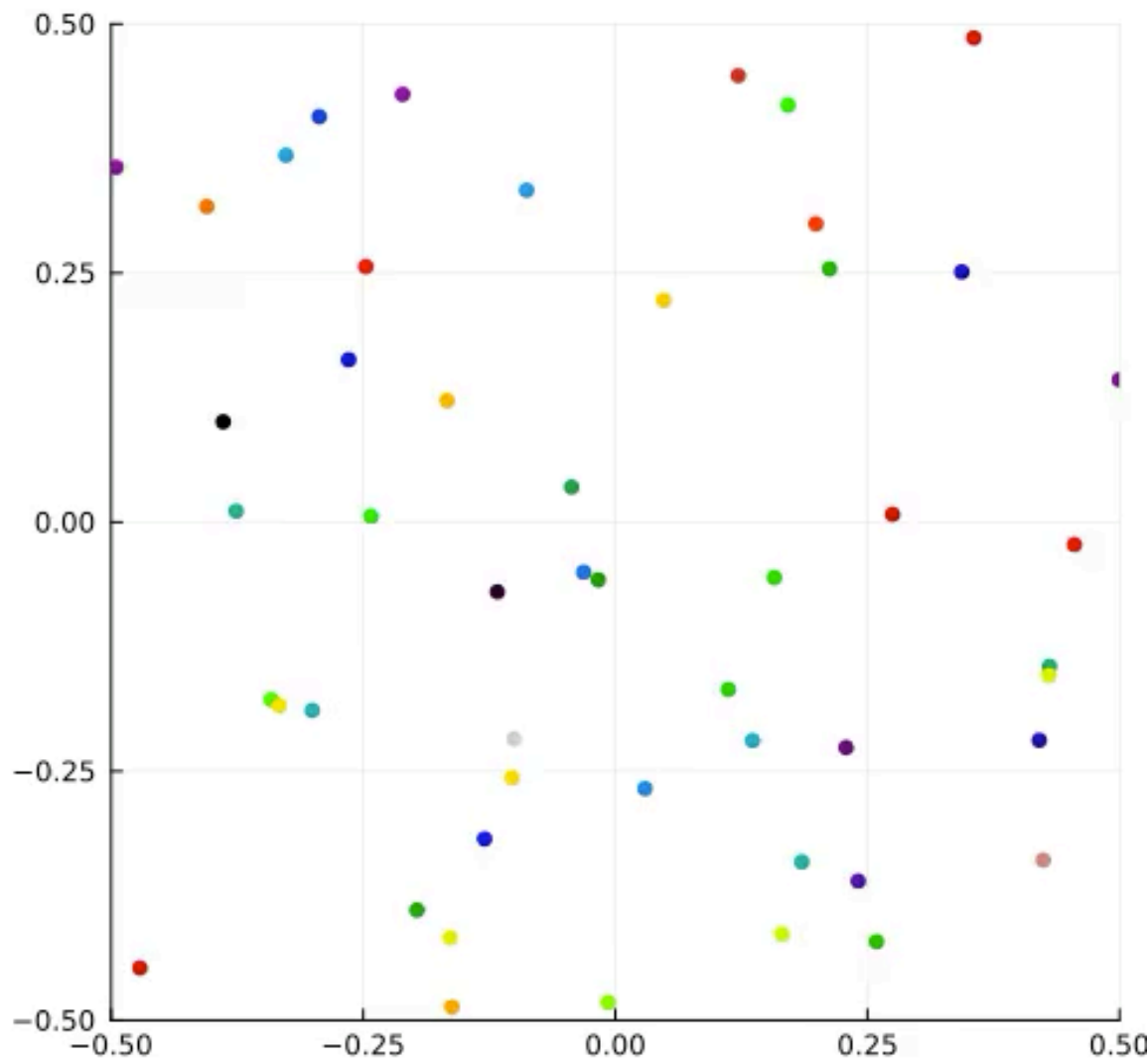
$t = 3$

ρ

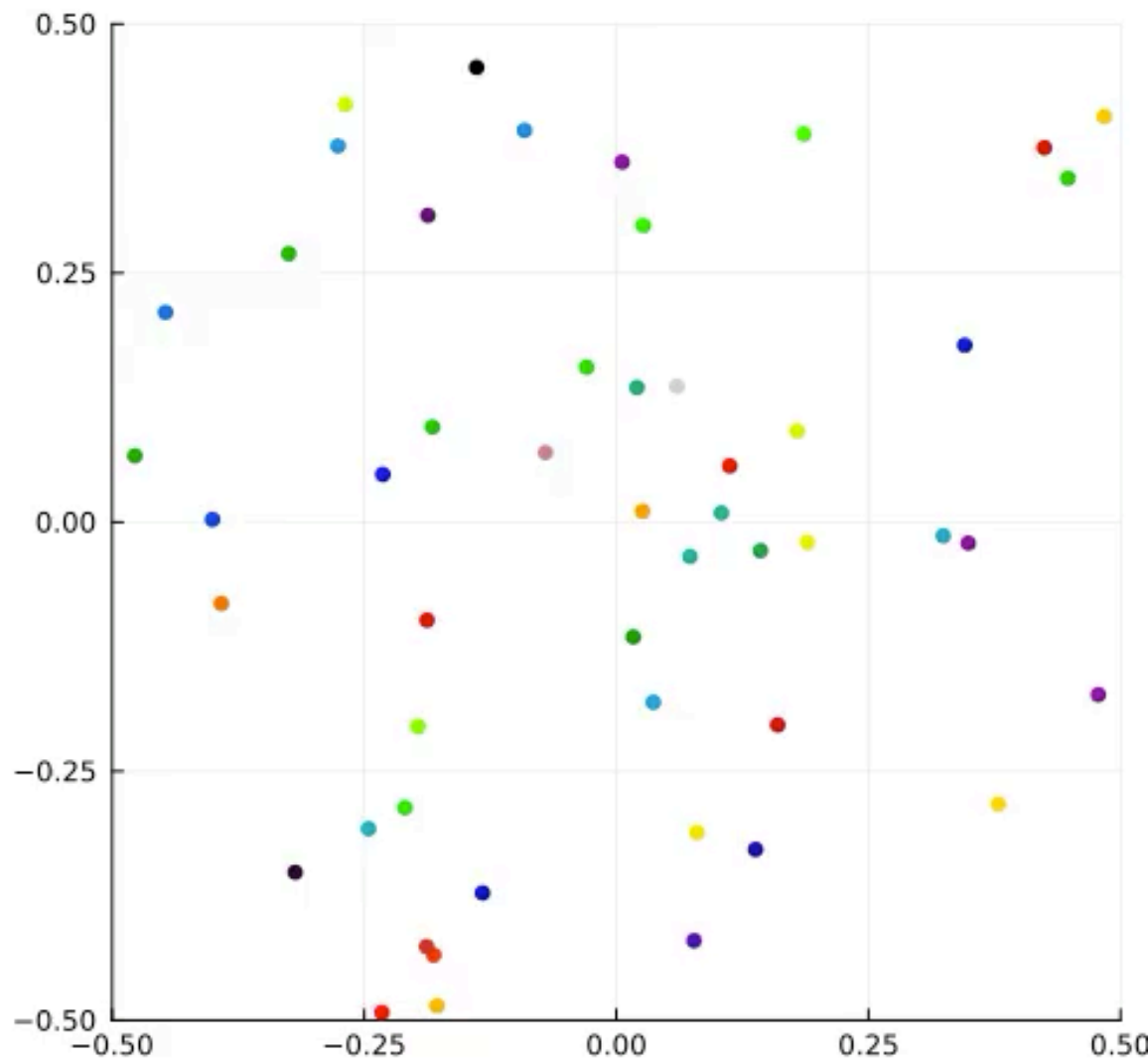


Antdynamics

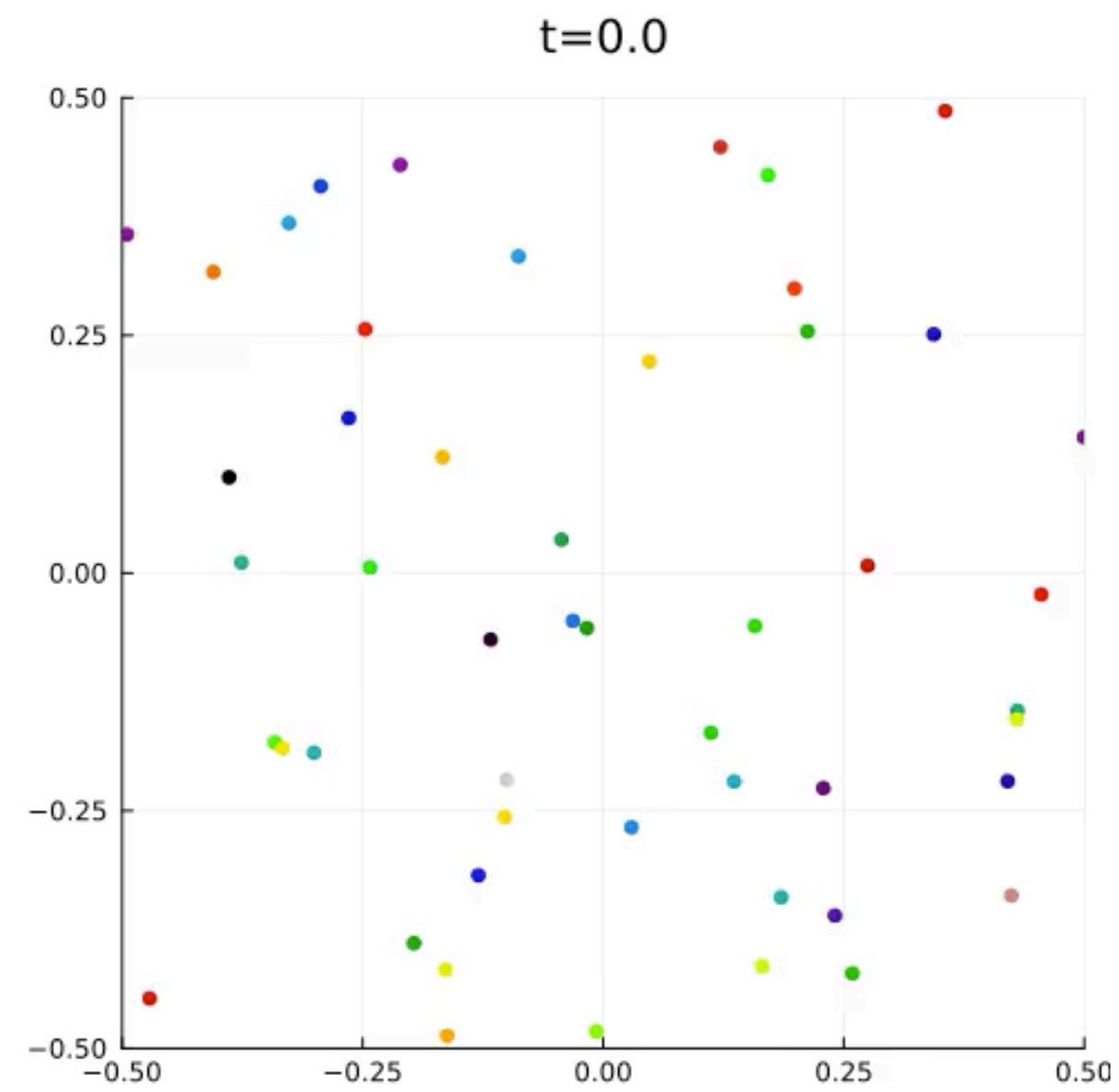
$t=0.0$



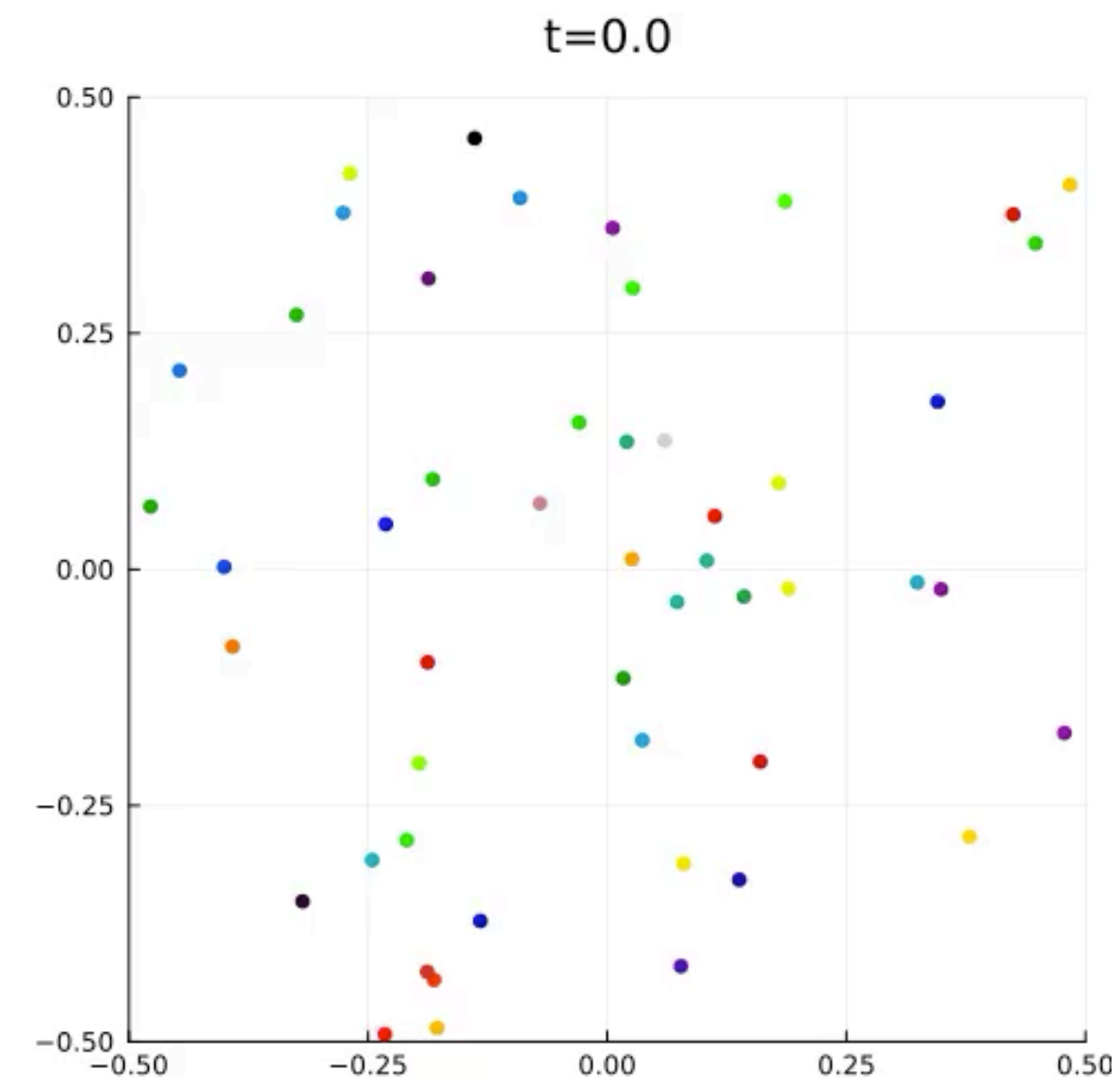
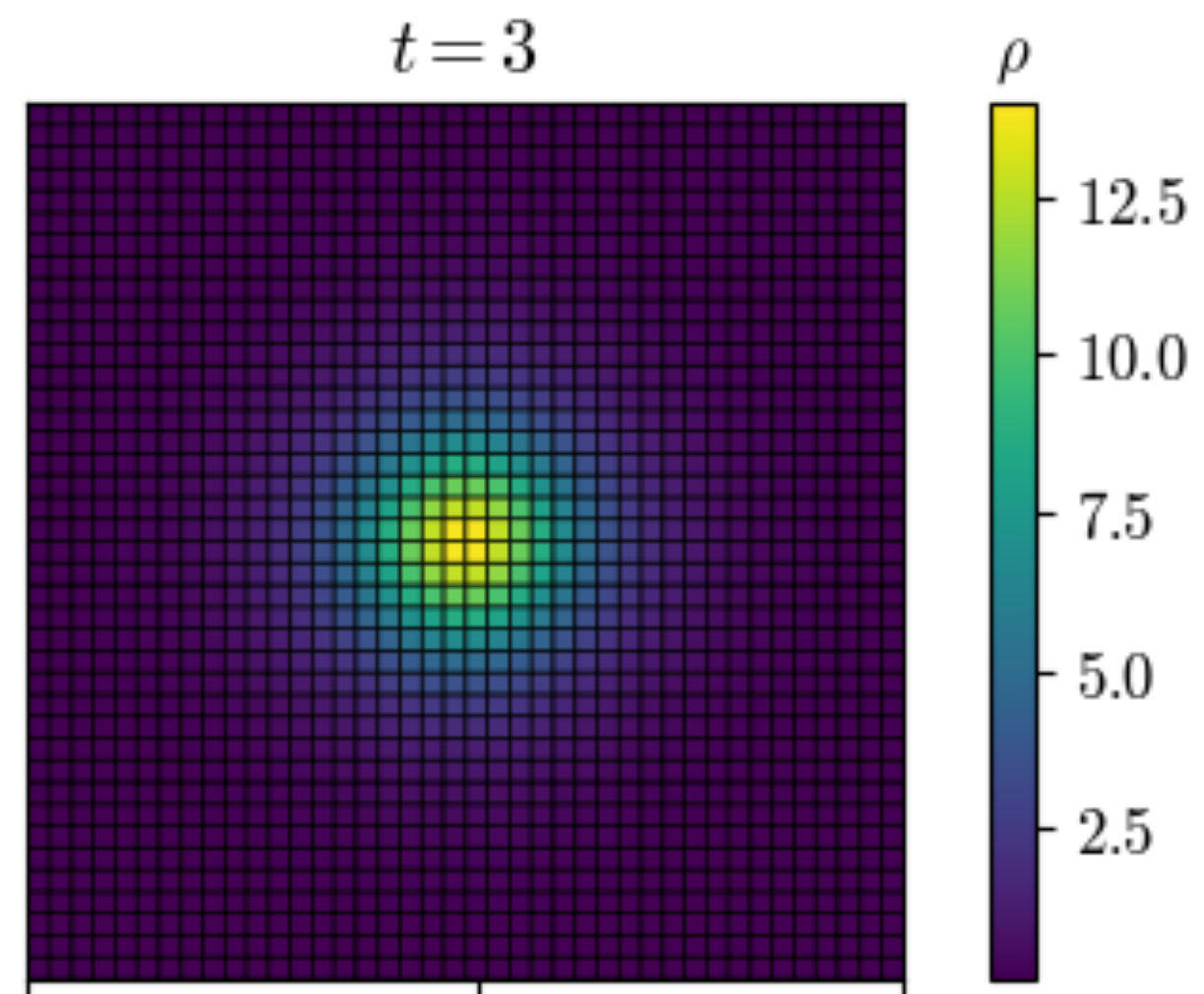
$t=0.0$



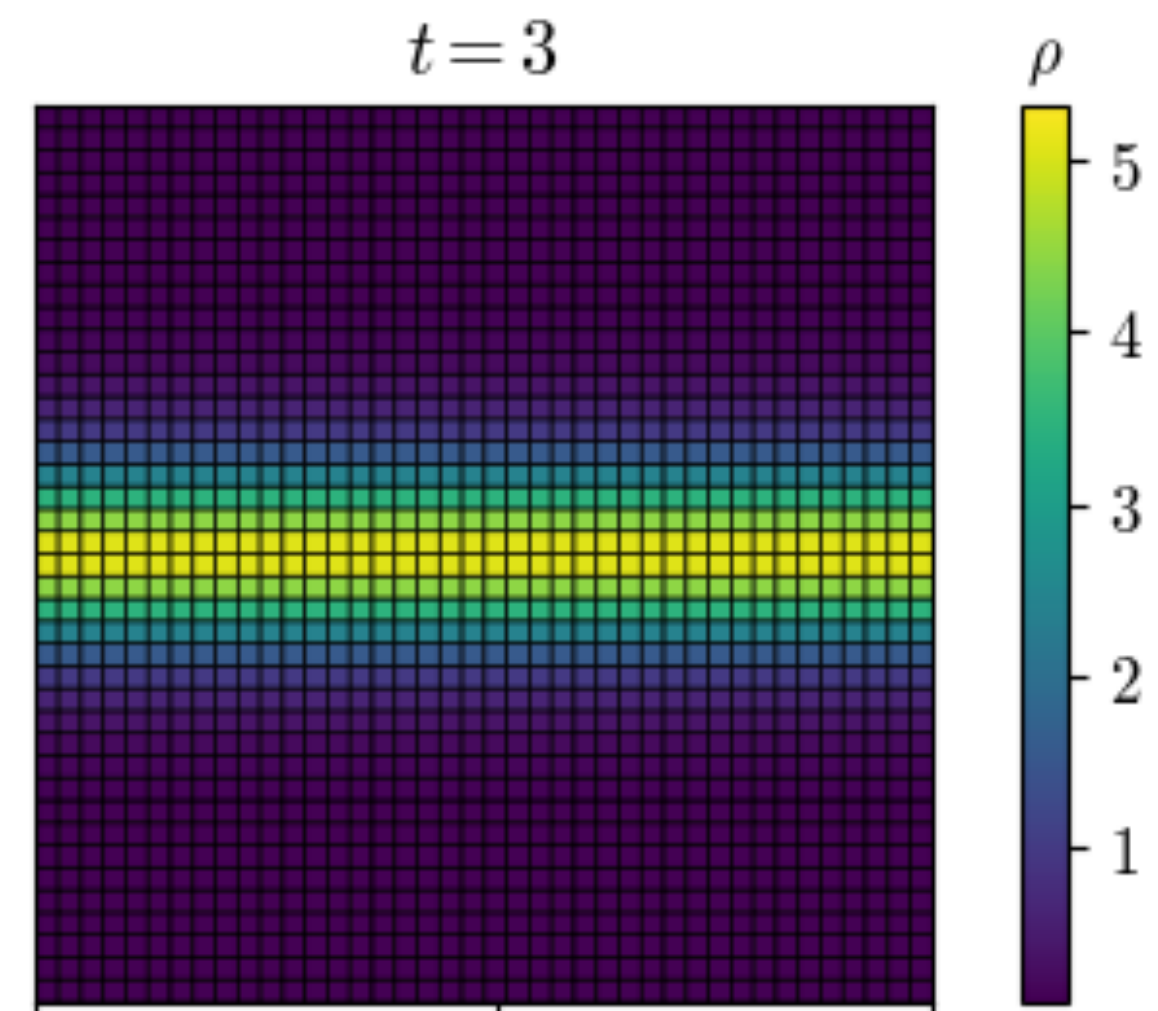
Ant dynamics



$$\tau = 0$$



$$\tau = 0.15$$



Ant dynamics

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) &= f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?
- Part 1 (Rakotomalala & dW, in review, 2025):

